

Autodesk® Scaleform®

ActionsScript 2 XML

Scaleform 4.4 ActionScript 2 XML

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: 3.0
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Autodesk® Scaleform® 4.4

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Autodesk® Scaleform® SDK
 PC
 Flash® Studio
 Scaleform
 Flash®
 Adobe®

Flash AS2 XML API XML XML XMLNode
 DOM XML
 DOM
 Scaleform Player SAX2 Scaleform XML
 Windows®, Linux®, MacOS®, Xbox 360®, PSP® (PlayStation® Portable), PlayStation®2 (PS2), PlayStation®3 (PS3™), Nintendo® Wii
 Scaleform XML

1. XML

Scaleform XML

1.1 ActionScript XML

ActionScript XML
FxPlayer.cpp

Scaleform

XML

```
...
#include "XML/XML_Expatri.h"
using namespace Scaleform;
...
Ptr<XML::Parser> pexpatXmlParser = *new XML::ParserExpatri;
Ptr<XML::SupportBase> pxmlSupport = *new XML::Support(pexpatXmlParser);
mLoader.SetXMLSupport(pxmlSupport);...
```

1.2 C++

XML

C++
DOM

XML
Scaleform

ActionScript XML

```
...
#include "GFx/XML/XML_DOM.h"
using namespace Scaleform;
...

//          DOM
XML::DOMBuilder domBuilder(Loader.GetXMLSupport());
//          DOM
XML::DOMBuilder domBuilder2(Loader.GetXMLSupport(), true);
...
//  xml      DOM
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                              mLoader.GetFileOpener());

//  xml      DOM
Ptr<XML::ObjectManager> pObjMgr = *new XML::ObjectManager();
Ptr<XML::Document> pdoc2 = domBuilder.ParseFile("inputfile.xml",
                                              mLoader.GetFileOpener(), pObjMgr);
...
```

ActionScript XML

```
...
#include "GFx/XML/XML_DOM.h"
#include "GFx/XML/XML_Expatri.h"
using namespace Scaleform;
...
Ptr<XML::Parser> pexpatXmlParser = *new XML::ParserExpatri;
```

```

Ptr<XML::Support> pxmlSupport = *new XML::Support(pexpatXmlParser);
...

//          DOM
XML::DOMBuilder domBuilder(pxmlSupport);
//          DOM
XML::DOMBuilder domBuilder2(pxmlSupport, true);
...
//
...

```

1.3 C++ DOM

```

DOM                                     Scaleform
...
...
using namespace Scaleform;
...
void PrintDOMTree(XML::Node* proot)
{
    switch (proot->Type)
    {
        case XML::ElementNodeType:
        {
            XML::ElementNode* pnode =
                static_cast< XML::ElementNode*>(proot);
            if (pnode->Prefix.GetSize() > 0)
            {
                printf("ELEM - '%s:%s' ns:'%s' prefix:'%s'"
                    " localname: '%s'",
                    pnode->Prefix.ToCStr(),
                    pnode->Value.ToCStr(),
                    pnode->Namespace.ToCStr(),
                    pnode->Prefix.ToCStr(),
                    pnode->Value.ToCStr());
            }
            else
            {
                printf("ELEM - '%s' ns:'%s' prefix:''"
                    " localname: '%s'",
                    pnode->Value.ToCStr(),
                    pnode->Namespace.ToCStr(),
                    pnode->Value.ToCStr());
            }
            for (XML::Attribute* attr = pnode->FirstAttribute;
                attr != NULL; attr = attr->Next)
            {
                printf(" {%s,%s}", attr->Name.ToCStr(),
                    attr->Value.ToCStr());
            }
            printf("\n");
            for (XML::Node* child = pnode->FirstChild; child != NULL;
                child = child->NextSibling)
                PrintDOMTree(child);
            break;

```

```

    }
    case TextNodeType:
    {
        printf("TEXT - '%s'\n", proot->Value.ToCStr());
        break;
    }
    default:
    {
        printf("UNKN\n");
    }
}
}
...
Ptr<XML::Document> pdoc = domBuilder.ParseFile("inputfile.xml",
                                              Loader.GetFileOpener());
PrintDOMTree(root);

```

1.4

Scaleform GfXMLConfig.h XML

XML DOM

```

//
#ifdef SF_BUILD_DEBUG
//          DOM
// #define SF_XML_DOCBUILDER_TRACE
//
//          DOM
// (
//
// #define SF_XML_DOCBUILDER_DOM_TREE_DUMP
//
#endif // #ifdef SF_BUILD_DEBUG

```

2.

Scaleform			XML
GFx::XML::Parser	GFx XML	XML_Support.h	
XML_Support.h	GFx::XML::ParserHandler	SAX2	

```

namespace Scaleform { namespace GFx { namespace XML {
//
// XML
//
//      /
//
// XML
//
class Parser : public RefCountBaseNTS<Parser, Stat_Default_Mem>
{
public:
    virtual ~Parser() {}

    //
    // Parse methods
    //
    virtual bool    ParseFile(const char* pfilename, FileOpenerBase* pfo,
                             ParserHandler* pphandler) = 0;
    virtual bool    ParseString(const char* pdata, UPInt len,
                               ParserHandler* pphandler) = 0;
};
}}} //SF::GFx::XML

```

XML	DOM	GFx::XML::ParserLocator
ParseFile	ParseString	
GFx::XM::LParserHandler		

```

namespace Scaleform { namespace GFx { namespace XML {
// SAX2
// DOM          SAX2
//
//
//
class ParserHandler : public RefCountBase<ParserHandler, StatMV_XML_Mem>
{
public:
    ParserHandler() {}
    virtual ~ParserHandler() {}

    //
    virtual void    StartDocument() = 0;
    virtual void    EndDocument() = 0;

    //
    virtual void    StartElement(const StringRef& prefix,

```

```

                                const StringRef& localname,
                                const ParserAttributes& atts) = 0;
virtual void      EndElement(const StringRef& prefix,
                                const StringRef& localname) = 0;
//
virtual void      PrefixMapping(const StringRef& prefix,
                                const StringRef& uri) = 0;
//
virtual void      Characters(const StringRef& text) = 0;
//
virtual void      IgnorableWhitespace(const StringRef& ws) = 0;
//
virtual void      SkippedEntity(const StringRef& name) = 0;

// Gfx::XMLParser                                Gfx::XML::ParserHandler
// // //                                xml
virtual void      SetDocumentLocator(const ParserLocator* plocator) = 0;
//
virtual void      Comment(const StringRef& text) = 0;
//
virtual void      Error(const ParserException& exception) = 0;
virtual void      FatalError(const ParserException& exception) = 0;
virtual void      Warning(const ParserException& exception) = 0;
};
}}} //SF::Gfx::XML

```

3.

3.1 ActionScript

XML

ActionScript	XMLNode	XML		
		ActionScript		DOM
		DOM		ActionScript
XML.load			XMLNode	DOM
ActionScript	XMLNode	DOM		
		XML	XML	XMLNode

```
//      DOM
XMLNode temp = root.firstChild;
//
temp.someProperty = someValue;
//
XMLNode temp = temp.nextSibling;
//
temp = temp.prevSibling;
//
trace(temp.someProperty)
```

Flash™		Scaleform		Scaleform
XMLNode				

	XMLNode		ActionScript
XMLNode			

```
//      DOM
XMLNode temp = root.firstChild;
//
temp.attributes.someProperty = someValue;
//
XMLNode temp = temp.nextSibling;
//
temp = temp.prevSibling;
//
trace(temp.attributes.someProperty)
```

Scaleform

3.2 *ActionScript XML.status*

ActionScript 2.0 XML

ActionScript

- 0
- -2 A CDATA
- -3 XML
- -4 DOCTYPE
- -5
- -6 XML
- -7
- -8
- -9
- -10

ActionScript

```
//  
// XML_ERROR_NONE = 0  
// XML_ERROR_INVALID_TOKEN = 0 (ie: XML.parse("http://www.google.com"))  
// XML_ERROR_UNCLOSED_CDATA_SECTION = -2  
// XML_ERROR_UNCLOSED_TOKEN = -3  
// XML_ERROR_INVALID_TOKEN = -4  
// XML_ERROR_INVALID_TOKEN = -5  
// XML_ERROR_UNCLOSED_TOKEN = -8  
// XML_ERROR_NO_ELEMENTS = -9  
// XML_ERROR_TAG_MISMATCH = -10  
//
```

GfX::XML::DOMBuilder

XML

GfX::XML::ParserLocator

3.3 *ActionScript XML.docTypeDecl*

Flash®	“ ActionScript XML		DOCTYPE
XML.docTypeDecl	DTD	”	Flash Scaleform